

TL;DR — highest ROI first

You're at **0.3564 vs 0.40**, so you need **~+0.044 LB**. I'd spend the next 1–2 submission days on **external sequence priors / pretraining**, not video/sensor-heavy datasets.

Rank	Recommendation	Why ROI is high	Expected lift	1-hour kill criterion
1	Pretrain a rally-sequence encoder on ShuttleSet, then fine-tune on TT	ShuttleSet is public, stroke-level, structured, turn-based, with shot type, landing coords, player/opponent locations, rally scores/outcomes; much closer to your competition table format than video datasets. ^{1 2}	+0.010 to +0.030 LB if current model lacks external sequence pretraining	If a quick frozen-encoder / synthetic-feature version gives < +0.003 CV or worsens calibration after 1 seed, kill
2	Inject external "next-shot prior" features from ShuttleSet + OpenTT/TTSwing priors into your existing GBDT/NN ensemble	Lower effort than full transfer learning: create probability/count features such as <code>P(next_action</code>	<code>prev_action,</code> <code>prev_zone,</code> <code>score_state,</code> <code>hand-side proxy)</code> and <code>P(rally_win</code>	<code>prefix)';</code> ShuttleSet has 36,492 strokes / 3,685 rallies, and Extended OpenTT adds frame-accurate stroke + point- +0.006 to +0.020 LB If offline OOF gain is < +0.002 after adding only Markov / prefix priors, kill

¹<https://github.com/wywyWang/CoachAI-Projects/tree/main/ShuttleSet>

²<https://arxiv.org/abs/2306.04948>

				outcome labels. ^{3 4}		
3	Add a small dual-head Transformer/GRU teacher and blend only if decorrelated	A 2026 TT stroke-forecasting paper reports dual encoder-decoder Transformer results of 57.2% top-1 stroke type, 42.8% top-1 landing zone, 98.2% / 91.8% top-5 , using stroke-level event streams. ⁵	+0.004 to +0.015 LB from blend/diversity, not as sole model	If correlation with your best model is >0.98 and OOF gain < +0.002 , kill		

1. Top 3 actionable recommendations

1) ShuttleSet transfer learning — best ROI

Resource:

- GitHub dataset: wywyWang/CoachAI-Projects/ShuttleSet — [ShuttleSet GitHub](#)
- Paper: **arXiv:2306.04948**, *ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis* — [arXiv](#)

Why this matters: ShuttleSet contains **104 sets, 3,685 rallies, 36,492 strokes, 44 matches, 27 top-ranking players**, with stroke type, landing area/coordinates, scorer, player location, opponent location, scores, and rally outcome fields.^{6 7} That is structurally much closer to a

³<https://github.com/wywyWang/CoachAI-Projects/tree/main/ShuttleSet>

⁴<https://arxiv.org/abs/2512.19327>

⁵https://www.mjssm.me/clanci/MJSSM_March_2026_Wu.pdf

⁶<https://github.com/wywyWang/CoachAI-Projects/tree/main/ShuttleSet>

⁷<https://arxiv.org/abs/2306.04948>

table-tennis tactical sequence competition than TTSwing sensor data or video-only table-tennis datasets.^{8 9}

Implementation steps — 4–6h

1. Map ShuttleSet schema to TT schema

- ball_round → strikeNumber
 - type → external shot/action token
 - landing_area, landing_x, landing_y → external landing-zone token / continuous coords
 - roundscore_A/B → score-state features
 - player, getpoint_player → point-winner label
 - player_location_*, opponent_location_* → positional context
- ShuttleSet explicitly provides these fields in its CSV structure.¹⁰

2. Pretrain a compact sequence model

- Input: previous strokes in rally.
- Tasks:
 - next action / shot type
 - next landing area
 - rally winner / point winner
- Architecture: 2-layer GRU or tiny Transformer, not huge.
- Keep embeddings small: 32–96 dim.
- Use masked prefix prediction: train on prefixes of length 1...N.

3. Fine-tune on TT train

- Replace output heads with TT labels: actionId, pointId, serverGetPoint.
- Either:
 - freeze encoder for 1–2 epochs then unfreeze, or
 - use pretrained embeddings only.

4. Blend, don't trust alone

- Add predictions as meta-features to your current best model.
- Blend with your strongest CV-stable models.
- Use per-target weights; external pretraining may help actionId/pointId more than serverGetPoint.

Expected lift

- **Likely:** +0.010 to +0.020 LB.

⁸<https://github.com/wywyWang/CoachAI-Projects/tree/main/ShuttleSet>

⁹<https://github.com/DEPhantom/TTSWING>

¹⁰<https://github.com/wywyWang/CoachAI-Projects/tree/main/ShuttleSet>

- **Upside:** +0.030 if your current model is mostly LightGBM/CatBoost with weak sequential representation.
- **Risk:** badminton tactical priors may misalign with table tennis shot taxonomy; mitigate by using encoder features, not direct class mapping.

1-hour kill criterion

After one quick run:

- If frozen ShuttleSet encoder features improve OOF by $< +0.003$ on your local CV, kill full fine-tune.
 - If per-target improvement appears only on one fold and not others, kill or use only as low-weight blend.
 - If model improves CV but public LB drops on 1 submission, don't spend second submission unless CV split is clearly leakage-safe.
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2) External prior features — fastest useful external-data path

Resources:

- ShuttleSet GitHub / paper for stroke-level priors.^{11 12}
- Extended OpenTT Games Dataset: **arXiv:2512.19327**, revised Jan 8, 2026, adds frame-accurate stroke annotations, posture labels, feet-position labels, and point outcome tags to OpenTTGames.^{13 14}
- TTSwing Dryad/GitHub for player-skill / swing-mode priors, but weaker match-sequence relevance.^{15 16}

Implementation steps — 2–4h

Build **external prior tables**, then merge as features by current rally prefix pattern. Do **not** use player ID mapping.

Suggested priors:

¹¹<https://github.com/wywyWang/CoachAI-Projects/tree/main/ShuttleSet>

¹²<https://arxiv.org/abs/2306.04948>

¹³<https://arxiv.org/abs/2512.19327>

¹⁴<https://www.alphaxiv.org/overview/2512.19327v1>

¹⁵<https://datadryad.org/dataset/doi:10.5061/dryad.0zpc8677f>

¹⁶<https://github.com/DEPhantom/TTSWING>

1. First-order / second-order transition priors

- $P(\text{next_action} \mid \text{prev_action})$
- $P(\text{next_action} \mid \text{prev_action}, \text{prev_landing_zone})$
- $P(\text{next_landing_zone} \mid \text{prev_action}, \text{prev_zone})$
- $P(\text{point_win} \mid \text{serve/receive}, \text{score_diff_bucket}, \text{prefix_len})$

2. Prefix-length priors

- prefix_len
- $P(\text{rally_ends_next} \mid \text{prefix_len}, \text{score_state})$
- $P(\text{server_wins} \mid \text{prefix_len}, \text{previous_action_group})$

3. Location dynamics

- ShuttleSet has landing and player/opponent location coordinates, so derive coarse features like:
 - previous landing side
 - distance between player and opponent
 - attack-depth proxy
 - zone-switch indicator
- ShuttleSet includes landing coordinates and player/opponent location coordinates per stroke. ¹⁷

4. Smoothing

- Use Bayesian smoothing:
 - global prior
 - conditional prior
 - count feature
- Add count/confidence as a separate feature so model can learn when to trust prior.

5. Train with current model

- Add these to CatBoost/LightGBM/NN.
- Use target-specific versions.
- For NN, concatenate prior logits to final head input.

Expected lift

- **Likely:** +0.006 to +0.015.
- **Upside:** +0.020 if your model lacks strong sequence-count features.
- Very good use of one LB submission because the feature impact should show quickly in OOF.

1-hour kill criterion

- Build only 3 priors first:

¹⁷<https://github.com/wywyWang/CoachAI-Projects/tree/main/ShuttleSet>

- a. $P(\text{action} \mid \text{prev_action}, \text{prefix_len_bucket})$
 - b. $P(\text{point} \mid \text{prev_zone}, \text{prefix_len_bucket})$
 - c. $P(\text{serverGetPoint} \mid \text{score_diff_bucket}, \text{prefix_len_bucket})$
 - If these give $< +0.002$ OOF, kill or keep only for low-weight blend.
 - If CV improves but calibration/entropy worsens heavily, keep priors only as meta-model features.
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3) Small TT-specific sequence teacher + blend

Resources:

- 2026 TT stroke forecasting paper: *A New Table Tennis Match Stroke Forecasting Method Using Transformer-Based Deep Neural Networks*, reports **57.2% top-1 stroke type, 42.8% top-1 landing zone, 98.2% top-5 stroke type, 91.8% top-5 landing zone** on a stroke-level Lin Yun-Ju match dataset. ¹⁸
- Public repo found for the same 2026 AI CUP-style task uses multi-output LSTM for actionId, pointId, and serverGetPoint, but it looks like a baseline repo rather than a proven top solution. ¹⁹

Implementation steps — 4–8h

1. Build a **tiny dual-head model**
 - Shared sequence encoder: GRU/Transformer.
 - Heads:
 - action
 - point
 - serverGetPoint
 - Add score features and previous stroke embeddings at every step.
2. Use **teacher outputs as blend features**
 - Don't rely on it as your final model unless CV is strong.
 - Export OOF logits/probabilities.
 - Feed into your best tree/stacking model.
3. Add **auxiliary losses**
 - Predict next action and next point.
 - Predict rally winner.
 - Optional: predict whether next stroke ends rally.
4. Keep model small
 - 2 layers max.

¹⁸https://www.mjssm.me/clanci/MJSSM_March_2026_Wu.pdf

¹⁹<https://github.com/lqi234488/AICUP-2026-Table-Tennis-Prediction>

- 64–128 hidden.
- 5 seeds only if first seed works.

Expected lift

- **Likely:** +0.004 to +0.010.
- **Upside:** +0.015 if current ensemble lacks deep sequence diversity.
- **Main value:** decorrelated logits for blending.

1-hour kill criterion

- Train 1 seed for 10–15 epochs.
 - If OOF is worse than current NN/GBDT and prediction correlation with current best is **>0.98**, kill.
 - If it helps only one target, use only that target's logits.
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2. Q1 — public table-tennis stroke-level datasets found

Found: TTSwing — public TT stroke/swing-level sensor dataset

Download / source:

- Dryad dataset: **TTSwing: Table Tennis Swing and racket kinematics analysis**, DOI 10.5061/dryad.0zpc8677f, with TTSWING.csv around 44 MB. ²⁰
- GitHub: DEPhantom/TTSWING, includes dataset and experiment code. ²¹
- Scientific Data 2025 paper: 93 elite Taiwanese players, 9-axis sensors embedded in custom racket grips, anonymized demographics. ²²

Usefulness for your competition: low-to-medium. It is true table tennis and stroke-level, but it is **racket kinematics / swing-mode data**, not rally tactical sequence data. ^{23 24} Use it only for weak priors such as skill/demographic/swing-mode embeddings if your features include action/spin/position but not actual IMU signals.

Kill fast: if adding TTSwing-derived priors gives **< +0.001–0.002 CV**, drop it.

²⁰<https://datadryad.org/dataset/doi:10.5061/dryad.0zpc8677f>

²¹<https://github.com/DEPhantom/TTSWING>

²²<https://www.nature.com/articles/s41597-025-04680-y>

²³<https://github.com/DEPhantom/TTSWING>

²⁴<https://datadryad.org/dataset/doi:10.5061/dryad.0zpc8677f>

Found: Extended OpenTT Games — public-ish TT video annotation dataset

Resource:

- **arXiv:2512.19327**, *Extended OpenTT Games Dataset: A table tennis dataset for fine-grained shot type and point outcome*, revised Jan 8, 2026.²⁵
- It extends OpenTTGames with frame-accurate shot type annotations, posture labels, feet-position labels, and point outcome tags.^{26 27}

Usefulness: medium if you can access annotations quickly; high only if your competition features can be aligned to shot type / posture / outcome. It is more TT-relevant than ShuttleSet, but less immediately plug-and-play if the release path requires digging through OpenTTGames/video assets.

Kill fast: if you can't get usable CSV/JSON annotations within **45 minutes**, do not burn your 4–8h block.

Found: OpenNeuro Real World Table Tennis — hit-event timing, IMU/EEG/video

Resource:

- OpenNeuro/GitHub dataset ds004505, includes EEG, EMG, IMU acceleration, MR images, and video from 25 participants; for 17 participants, video data includes timing of hit events marked in Adobe Premiere project files.²⁸

Usefulness: low for Kaggle tactical tabular prediction. It has TT hit events, but the modality mismatch is large.

²⁵<https://arxiv.org/abs/2512.19327>

²⁶<https://arxiv.org/abs/2512.19327>

²⁷<https://www.alphaxiv.org/overview/2512.19327v1>

²⁸<https://github.com/OpenNeuroDatasets/ds004505>

If ShuttleSet is treated as the closest match: transfer protocol

Yes — for **rally-sequence prediction**, ShuttleSet is the closest practical public dataset I found because it is structured, stroke-level, and includes tactical context and rally outcomes.^{29 30}

Protocol I'd use:

1. **Normalize both sports to abstract tokens**
 - action family: serve / attack / defensive / placement / transition
 - location: near/mid/deep, left/middle/right
 - rally state: prefix length, score diff, server/receiver role
2. **Pretrain masked next-stroke model**
 - Predict next shot family, next landing zone, rally winner.
 - Train on all ShuttleSet rally prefixes.
3. **Discard ShuttleSet output heads**
 - Keep only embeddings / encoder.
 - Reinitialize TT-specific heads.
4. **Fine-tune on TT**
 - Lower LR for encoder.
 - Higher LR for heads.
 - Early stop on your trusted CV.
5. **Use as blend feature**
 - Export OOF logits.
 - Stack/blend with current best.

This avoids brittle badminton-to-table-tennis label mapping while keeping turn-based tactical structure.

3. Q2 — 2024–2026 papers / writeups worth using

Table tennis / racket-sport papers

1. **Extended OpenTT Games Dataset**
 - **arXiv:2512.19327**, submitted Dec 22, 2025, revised Jan 8, 2026.³¹

²⁹<https://github.com/wywyWang/CoachAI-Projects/tree/main/ShuttleSet>

³⁰<https://arxiv.org/abs/2306.04948>

³¹<https://arxiv.org/abs/2512.19327>

- Adds fine-grained stroke annotations and point-outcome labels to OpenTTGames.^{32 33}
 - Actionability: use for external shot/outcome priors if data files are accessible fast.
2. **A New Table Tennis Match Stroke Forecasting Method Using Transformer-Based Deep Neural Networks**
- Published/early access around late 2025 / final 2026 in *Montenegrin Journal of Sports Science and Medicine*.³⁴
 - Reported metrics: **57.2% top-1 stroke type, 42.8% top-1 landing zone, 98.2% top-5 stroke type, 91.8% top-5 landing zone**.³⁵
 - Actionability: copy the **dual encoder-decoder / rally + player-context fusion idea**, not necessarily the full architecture.
3. **TTSwing Scientific Data 2025**
- Public TT swing/racket-kinematics dataset from 93 elite Taiwanese players with 9-axis racket-grip sensors.^{36 37}
 - Actionability: low unless you need TT-specific swing priors; don't prioritize above ShuttleSet.
4. **ShuttleSet — KDD 2023 but still most useful external data**
- **arXiv:2306.04948**; 36,492 stroke-level badminton records with shot type, landing location, player/opponent locations, and rally outcome.^{38 39}
 - Actionability: strongest external pretraining source despite being badminton.

³²<https://arxiv.org/abs/2512.19327>

³³<https://www.alphaxiv.org/overview/2512.19327v1>

³⁴https://www.mjssm.me/clanci/MJSSM_March_2026_Wu.pdf

³⁵https://www.mjssm.me/clanci/MJSSM_March_2026_Wu.pdf

³⁶<https://www.nature.com/articles/s41597-025-04680-y>

³⁷<https://datadryad.org/dataset/doi:10.5061/dryad.0zpc8677f>

³⁸<https://arxiv.org/abs/2306.04948>

³⁹<https://github.com/wywyWang/CoachAI-Projects/tree/main/ShuttleSet>

Kaggle / competition-method references

- NVIDIA's 2025 Kaggle Grandmasters playbook emphasizes fast experimentation, careful validation, diverse baselines, feature engineering, hill climbing, stacking, pseudo-labeling, and extra training for tabular competitions. ⁴⁰
 - Public Kaggle solution indexes exist and are useful for quickly finding ensemble/stacking patterns, but I did **not** find a high-quality public winning writeup specifically for this table-tennis competition. ^{41 42}
 - A public GitHub repo for **2026 AI CUP table tennis prediction** uses a multi-output LSTM for actionId, pointId, and serverGetPoint, but it appears baseline-level, not a leaderboard-winning writeup. ⁴³
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What I would actually do in your next 8 hours

Submission 1 candidate — external prior features

1. Download ShuttleSet.
2. Build 5–10 smoothed prior features.
3. Add to current best model.
4. Submit only if OOF improves $\geq +0.003$.

Submission 2 candidate — ShuttleSet pretrained encoder

1. Pretrain tiny GRU/Transformer on ShuttleSet.
2. Fine-tune on TT.
3. Export logits.
4. Blend with current best.
5. Submit only if OOF improves $\geq +0.005$ or the model is clearly decorrelated.

Submission 3 candidate — conservative blend

Blend:

- current best submission

⁴⁰<https://developer.nvidia.com/blog/the-kaggle-grandmasters-playbook-7-battle-tested-modeling-techniques-for-tabular-data/>

⁴¹https://github.com/anui0456/kaggle_competition_solutions

⁴²<https://liran1460.github.io/kaggle-solutions/>

⁴³<https://github.com/lqi234488/AICUP-2026-Table-Tennis-Prediction>

- external-prior model
- pretrained-sequence model

Use hill-climb weights on OOF; don't overweight the external model unless CV/LB confirms.

My sharp recommendation: **skip TTSwing unless you have spare time**. Use **ShuttleSet first**, because it is the only immediately usable, public, stroke-level, tactical sequence dataset with the right structure.